

TUM-RC coefficients

Oksana Chernova

29 January, 2026

Contents

Loading libraries	1
Read data	1
Missing values	2
Model for complete cases	2
Model without priornegbiopsy	3
Model without DRE	3
Model without volume	4
Model without priornegbiopsy and DRE	4
Model without priornegbiopsy and volume	5
Model without volume and DRE	5
Model without DRE, volume, priornegbiopsy	6

This RMarkdown file contains the TUM-RC model coefficients used to predict clinically significant prostate cancer (defined as ISUP ≥ 2).

Loading libraries

```
library(knitr)
library(dplyr)
library(tidyr)
library(stringr)
```

```
## Loading required package: common
```

Read data

```
data <- read.csv("C:/Users/ochernova/mydocs/pr2/MRibiopsy21.01.2026.csv")
dim(data)
```

```
## [1] 1333    7
```

```

# Define clinically significant prostate cancer as ISUP >= 2
data <- data %>%
  mutate(
    PSA = as.numeric(psa),
    PIRADS = as.numeric(PIRADS),
    csPCa = as.factor(ifelse(ISUP >= 2, "csPCa", "non-csPCa")),
    csPCa2 = ifelse(ISUP >= 2, 1, 0),
    log2_psa = log2(psa),
    log2_vol = log2(volume)
  )

# Define factor levels ----
data <- data %>% mutate(
  priornegbiopsy = relevel(factor(priornegbiopsy), ref="no")
  ,DRE = relevel(factor(DRE), ref="normal")
)

data = data[,c("age", "log2_psa", "log2_vol", "PIRADS", "DRE", "priornegbiopsy", "csPCa2")]

```

Missing values

```

na_counts <- colSums(is.na(data))
print(na_counts)

```

```

##          age      log2_psa      log2_vol      PIRADS      DRE
##          0            0            81            0        27
## priornegbiopsy      csPCa2
##          1            0

```

Model for complete cases

```

data_complete <- data %>%
  filter(!is.na(DRE) & !is.na(log2_vol) & !is.na(PIRADS) & !is.na(priornegbiopsy))

print(paste0("sample size: ", nrow(data_complete)))

## [1] "sample size: 1233"

model1 <- glm(formula = csPCa2 ~ age + log2_psa + PIRADS + log2_vol + DRE + priornegbiopsy,
  data = data_complete,
  family=binomial(link='logit'))

sumreg(model1)

```

	Estimate	OR (95% CI)	P value
Intercept	-2.2792	0.10 (0.02, 0.45)	0.003
Age	0.0514	1.05 (1.03, 1.07)	< 0.001
log2 PSA	0.4622	1.59 (1.3, 1.94)	< 0.001
PIRADS	1.3143	3.72 (2.95, 4.7)	< 0.001
log2 Volume	-1.1906	0.30 (0.23, 0.4)	< 0.001
DRE abnormal	0.9155	2.50 (1.62, 3.84)	< 0.001

	Estimate	OR (95% CI)	P value
Prior negative biopsy Yes	-1.0907	0.34 (0.21, 0.54)	< 0.001

```
paste(as.character(round(summary(model1)$coef[, 1], 5)), sep="' '", collapse=", ")
```

```
## [1] "-2.27918, 0.05139, 0.46219, 1.31431, -1.19057, 0.91551, -1.0907"
```

Model without priornegbiopsy

```
data_bx <- data %>%
  filter(!is.na(DRE) & !is.na(log2_vol) & !is.na(PIRADS) )
```

```
print(paste0("sample size: ", nrow(data_bx)))
```

```
## [1] "sample size: 1233"
```

```
model2 <- glm(formula = csPCa2 ~ age + log2_psa + PIRADS + log2_vol + DRE ,
  data = data_complete,
  family=binomial(link='logit'))
```

```
sumreg(model2)
```

	Estimate	OR (95% CI)	P value
Intercept	-1.9988	0.14 (0.03, 0.59)	0.008
Age	0.0517	1.05 (1.03, 1.07)	< 0.001
log2 PSA	0.4002	1.49 (1.23, 1.81)	< 0.001
PIRADS	1.3145	3.72 (2.95, 4.69)	< 0.001
log2 Volume	-1.2380	0.29 (0.22, 0.38)	< 0.001
DRE abnormal	0.9398	2.56 (1.67, 3.92)	< 0.001

```
paste(as.character(round(summary(model2)$coef[, 1], 5)), sep="' '", collapse=", ")
```

```
## [1] "-1.99885, 0.05169, 0.4002, 1.31446, -1.23796, 0.93983"
```

Model without DRE

```
data_dre <- data %>%
  filter(!is.na(log2_vol) & !is.na(priornegbiopsy) & !is.na(PIRADS))
```

```
print(paste0("sample size: ", nrow(data_dre)))
```

```
## [1] "sample size: 1251"
```

```
model3 <- glm(formula = csPCa2 ~ age + log2_psa + PIRADS + log2_vol + priornegbiopsy,
  data = data_dre,
  family=binomial(link='logit'))
```

```
sumreg(model3)
```

	Estimate	OR (95% CI)	P value
Intercept	-2.7077	0.07 (0.02, 0.29)	< 0.001
Age	0.0555	1.06 (1.04, 1.08)	< 0.001
log2 PSA	0.4465	1.56 (1.29, 1.89)	< 0.001
PIRADS	1.4371	4.21 (3.35, 5.29)	< 0.001
log2 Volume	-1.2031	0.30 (0.23, 0.39)	< 0.001
Prior negative biopsy Yes	-1.0804	0.34 (0.21, 0.54)	< 0.001

```
paste(as.character(round(summary(model3)$coef[, 1], 5)), sep="' '", collapse=", ")
```

```
## [1] "-2.70773, 0.0555, 0.44646, 1.43709, -1.20305, -1.08037"
```

Model without volume

```
data_volume <- data %>%
  filter(!is.na(DRE) & !is.na(priornegbiopsy) & !is.na(PIRADS))

print(paste0("sample size: ", nrow(data_volume)))

## [1] "sample size: 1306"

model4 <- glm(formula = csPCa2 ~ age + log2_psa + PIRADS + DRE + priornegbiopsy ,
  data = data_volume,
  family=binomial(link='logit'))

sumreg(model4)
```

	Estimate	OR (95% CI)	P value
Intercept	-6.5952	0.00 (0, 0)	< 0.001
Age	0.0266	1.03 (1.01, 1.04)	< 0.001
log2 PSA	0.1999	1.22 (1.03, 1.45)	0.023
PIRADS	1.2956	3.65 (2.94, 4.54)	< 0.001
DRE abnormal	1.0767	2.94 (1.97, 4.38)	< 0.001
Prior negative biopsy Yes	-1.2503	0.29 (0.19, 0.44)	< 0.001

```
paste(as.character(round(summary(model4)$coef[, 1], 5)), sep="' '", collapse=", ")
```

```
## [1] "-6.59522, 0.02665, 0.19985, 1.29563, 1.07673, -1.25034"
```

Model without priornegbiopsy and DRE

```
data_bxdre <- data %>%
  filter( !is.na(log2_vol) & !is.na(PIRADS) )

print(paste0("sample size: ", nrow(data_bxdre)))

## [1] "sample size: 1252"

model5 <- glm(formula = csPCa2 ~ age + log2_psa + PIRADS + log2_vol,
  data = data_bxdre,
  family=binomial(link='logit'))
```

```
sumreg(model5)
```

	Estimate	OR (95% CI)	P value
Intercept	-2.4592	0.09 (0.02, 0.37)	< 0.001
Age	0.0553	1.06 (1.04, 1.08)	< 0.001
log2 PSA	0.3875	1.47 (1.22, 1.77)	< 0.001
PIRADS	1.4469	4.25 (3.39, 5.33)	< 0.001
log2 Volume	-1.2468	0.29 (0.22, 0.37)	< 0.001

```
paste(as.character(round(summary(model5)$coef[, 1], 5)), sep="' '", collapse=", ")
```

```
## [1] "-2.45922, 0.05533, 0.38753, 1.44691, -1.24684"
```

Model without priornegbiopsy and volume

```
data_bxvol <- data %>%
  filter(!is.na(DRE) & !is.na(PIRADS) )

print(paste0("sample size: ", nrow(data_bxvol)))
```

```
## [1] "sample size: 1306"
```

```
model6 <- glm(formula = csPCa2 ~ age + log2_psa + PIRADS + DRE,
  data = data_bxvol,
  family=binomial(link='logit'))
```

```
sumreg(model6)
```

	Estimate	OR (95% CI)	P value
Intercept	-6.4739	0.00 (0, 0)	< 0.001
Age	0.0260	1.03 (1.01, 1.04)	0.001
log2 PSA	0.1088	1.11 (0.95, 1.31)	0.194
PIRADS	1.3001	3.67 (2.96, 4.55)	< 0.001
DRE abnormal	1.0889	2.97 (2, 4.41)	< 0.001

```
paste(as.character(round(summary(model6)$coef[, 1], 5)), sep="' '", collapse=", ")
```

```
## [1] "-6.47392, 0.026, 0.10883, 1.30009, 1.08885"
```

Model without volume and DRE

```
data_volumedre <- data %>%
  filter(!is.na(priornegbiopsy) & !is.na(PIRADS))

print(paste0("sample size: ", nrow(data_volumedre)))
```

```
## [1] "sample size: 1332"
```

```
model7 <- glm(formula = csPCa2 ~ age + log2_psa + PIRADS + priornegbiopsy,
              data = data_volumedre,
              family=binomial(link='logit'))

sumreg(model7)
```

	Estimate	OR (95% CI)	P value
Intercept	-7.2284	0.00 (0, 0)	< 0.001
Age	0.0319	1.03 (1.02, 1.05)	< 0.001
log2 PSA	0.1808	1.20 (1.02, 1.41)	0.030
PIRADS	1.4468	4.25 (3.43, 5.26)	< 0.001
Prior negative biopsy Yes	-1.2398	0.29 (0.19, 0.44)	< 0.001

```
paste(as.character(round(summary(model7)$coef[, 1], 5)), sep="' '", collapse=", ")
```

```
## [1] "-7.22844, 0.03191, 0.18077, 1.44682, -1.23983"
```

Model without DRE, volume, priornegbiopsy

```
data_volumedrebx <- data %>%
  filter(!is.na(PIRADS))

print(paste0("sample size: ", nrow(data_volumedrebx)))
```

```
## [1] "sample size: 1333"
```

```
model8 <- glm(formula = csPCa2 ~ age + log2_psa + PIRADS,
              data = data_volumedrebx,
              family=binomial(link='logit'))

sumreg(model8)
```

	Estimate	OR (95% CI)	P value
Intercept	-7.1530	0.00 (0, 0)	< 0.001
Age	0.0312	1.03 (1.02, 1.05)	< 0.001
log2 PSA	0.0963	1.10 (0.94, 1.29)	0.226
PIRADS	1.4611	4.31 (3.49, 5.32)	< 0.001

```
paste(as.character(round(summary(model8)$coef[, 1], 5)), sep="' '", collapse=", ")
```

```
## [1] "-7.15298, 0.03116, 0.09626, 1.46109"
```